Complete Linux Guide — From Scratch to Advanced

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1. What is Linux?

Linux is an open-source operating system kernel created by Linus Torvalds.

Linux is used in:

- Servers
- Cloud computing
- Android phones
- Supercomputers
- DevOps
- Cybersecurity
- Embedded systems

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2. Linux Distributions (Distros)

Distro	Usage
Ubuntu	Beginner friendly
Debian	Stable servers
Fedora	Latest technologies
CentOS	Enterprise
Kali Linux	Ethical hacking
Arch Linux	Advanced users

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3. Linux Architecture

User
↓
Shell

Sai_Cloud_Services

↓
Kernel
↓
Hardware

Components

Kernel

Core of Linux.
Manages:

- CPU
- RAM
- Devices
- Processes

Shell

Interface between user and kernel.

Examples:

- Bash
- Zsh
- Fish

File System

Stores files and directories.

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4. Linux File System Structure

```
/
 ├── bin
 ├── boot
 ├── dev
```

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```
├── etc
├── home
├── lib
├── media
├── opt
├── proc
├── root
├── tmp
├── usr
└── var
```

Important Directories

Directory	Purpose
/	Root
/home	User files
/etc	Configurations
/var	Logs
/tmp	Temporary files
/usr	Programs
/root	Root user

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5. Basic Linux Commands

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pwd — Current Directory

pwd

Shows current path.

Example:

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/home/user

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ls — List Files

ls

Options

ls -l

Long format.

ls -a

Hidden files.

ls -lh

Readable sizes.

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cd — Change Directory

cd Documents

Back:

cd ..

Home:

cd ~

Root:

cd /

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mkdir — Create Directory

mkdir project

Nested:

mkdir -p a/b/c

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rmdir — Remove Empty Folder

rmdir test

---

touch — Create File

touch notes.txt

Multiple:

touch a.txt b.txt c.txt

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cat — View File

cat file.txt

Create file:

cat > notes.txt

Save:

CTRL + D

---

cp — Copy Files

cp file1 file2

Copy folder:

cp -r folder1 folder2

---

mv — Move/Rename

Rename:

mv old.txt new.txt

Move:

mv file.txt /home/user

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rm — Delete

Delete file:

rm file.txt

Delete folder:

rm -r folder

Force delete:

rm -rf folder

⚠ Dangerous command.

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clear — Clear Screen

clear

Shortcut:

CTRL + L

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echo — Print Text

echo "Hello"

Write to file:

echo "Linux" > file.txt

Append:

echo "Linux" >> file.txt

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6. Viewing Files

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head — First Lines

head file.txt

5 lines:

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head -5 file.txt

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tail — Last Lines

tail file.txt

Live logs:

tail -f logfile.log

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less — Read Large File

less file.txt

Quit:

q

---

more

more file.txt

---

7. Text Editors

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nano — Beginner Editor

nano file.txt

Save:

CTRL + O

Exit:

CTRL + X

---

vim — Advanced Editor

Vim

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vim file.txt

Commands:

i → insert
ESC → normal mode
:wq → save quit
:q! → force quit

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8. Searching Commands

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grep — Search Text

grep "hello" file.txt

Ignore case:

grep -i hello file.txt

Recursive:

grep -r hello folder

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find — Search Files

find /home -name file.txt

Find txt files:

find . -name "*.txt"

---

locate — Fast Search

locate file.txt

Update DB:

sudo updatedb

---

which — Find Command Path

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which python

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9. File Permissions

Linux permissions:

r = read
w = write
x = execute

Example:

-rwxr-xr--

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chmod — Change Permissions

chmod 755 script.sh

Permission math:

755=7+5+5755=7+5+5755=7+5+5

7 = rwx
5 = r-x
5 = r-x

---

chown — Change Owner

sudo chown john file.txt

---

chgrp — Change Group

sudo chgrp developers file.txt

---

10. Users & Groups

---

whoami

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whoami

---

id

id

---

adduser

sudo adduser john

---

passwd

passwd

---

su

su john

---

sudo

sudo apt update

---

11. Package Management

---

apt (Ubuntu/Debian)

Update:

sudo apt update

Upgrade:

sudo apt upgrade

Install:

sudo apt install nginx

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Remove:

sudo apt remove nginx

Search:

apt search nginx

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yum (CentOS)

sudo yum install nginx

---

dnf (Fedora)

sudo dnf install nginx

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12. Process Management

---

ps — Running Processes

ps aux

---

top — Live Monitoring

top

Quit:

q

---

htop

htop

---

kill — Stop Process

kill PID

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Force:

kill -9 PID

---

pkill — Kill by Name

pkill firefox

---

jobs — Background Jobs

jobs

---

fg — Bring Job Foreground

fg

---

bg — Run in Background

bg

---

13. Networking Commands

---

ip

ip addr

---

ping

ping google.com

---

ssh

ssh user@server

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scp

scp file.txt user@server:/home/user

---

wget

wget https://example.com/file.zip

---

curl

curl https://example.com

---

netstat

netstat -tulnp

---

ss

ss -tulnp

---

traceroute

traceroute google.com

---

nslookup

nslookup google.com

---

dig

dig google.com

---

14. Compression Commands

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tar

Create:

tar -cvf backup.tar folder

Extract:

tar -xvf backup.tar

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gzip

gzip file.txt

Unzip:

gunzip file.txt.gz

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zip

zip files.zip file1 file2

---

unzip

unzip files.zip

---

15. Disk Management

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df — Disk Usage

df -h

---

du — Folder Size

du -sh folder

---

lsblk — Disk Partitions

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lsblk

---

fdisk

sudo fdisk -l

---

mount

mount

---

umount

umount /dev/sdb1

---

16. System Monitoring

---

free — RAM

free -h

---

uptime

uptime

---

vmstat

vmstat

---

iostat

iostat

---

journalctl

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journalctl -xe

---

dmesg

dmesg

---

17. Environment Variables

View:

printenv

Home:

echo $HOME

Set:

export NAME=Linux

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18. Shell Scripting

First Script

```
#!/bin/bash

echo "Hello Linux"
```

Run:

```
chmod +x script.sh
./script.sh
```

Variables

```
name="Linux"
echo $name
```

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if Condition

```
if [ $a -gt $b ]
then
 echo "a greater"
fi
```

---

for Loop

```
for i in 1 2 3
do
 echo $i
done
```

---

while Loop

```
while true
do
 echo "running"
done
```

---

case Statement

```
case $var in
 start) echo "start";;
 stop) echo "stop";;
esac
```

Functions

```
hello() {
 echo "Hello"
}
```

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19. Cron Jobs

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Edit:

crontab -e

Example:

* * * * * /home/script.sh

Runs every minute.

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20. Services

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systemctl

Start:

sudo systemctl start nginx

Enable:

sudo systemctl enable nginx

Restart:

sudo systemctl restart nginx

Status:

systemctl status nginx

---

service

service nginx restart

---

21. SSH Authentication

Generate key:

ssh-keygen

Copy:

ssh-copy-id user@server

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22. Pipes & Redirection

Pipe

cat file.txt | grep hello

Redirect Output

Overwrite:

echo hi > file.txt

Append:

echo hi >> file.txt

---

stderr

command 2> error.txt

---

stdout + stderr

command > all.txt 2>&1

---

23. File Descriptors

0 = stdin
1 = stdout
2 = stderr

---

24. Linux Boot Process

BIOS/UEFI
↓
Bootloader

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↓
Kernel
↓
systemd/init
↓
Services

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25. Important Logs

File	Purpose
/var/log/syslog	System logs
/var/log/auth.log	Login logs
/var/log/kern.log	Kernel logs

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26. Firewall Security

UFW

Allow SSH:

sudo ufw allow 22

Enable:

sudo ufw enable

Status:

sudo ufw status

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27. Bash Shortcuts

Shortcut	Action
Ctrl + C	Stop process

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Shortcut	Action
Ctrl + Z	Suspend
Ctrl + L	Clear
Ctrl + R	Search history
Tab	Auto complete

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28. Important Linux Tools

Tool	Usage
Git	Version control
Docker	Containers
Kubernetes	Container orchestration
Ansible	Automation
Jenkins	CI/CD

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29. Linux Interview Topics

- Permissions
- Process vs Thread
- Hard link vs Soft link
- grep/sed/awk
- Networking
- SSH
- Cron jobs
- Shell scripting
- systemd

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30. Linux Learning Roadmap

Beginner

- Navigation
- File handling
- Permissions
- Users

Intermediate

- Networking
- Shell scripting
- Services
- SSH

Advanced

- Docker
- Kubernetes
- DevOps
- Security
- Automation

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31. Practice Tasks

Beginner

- Create/delete files
- Change permissions
- Search files

Intermediate

- Write shell scripts

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- Setup SSH
- Configure cron

Advanced

- Host web server
- Configure firewall
- Dockerize applications
- Automate backups

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32. Best Linux Learning Resources

- Linux Journey
- Linux Documentation Project
- Ubuntu Documentation
- Arch Wiki
- GNU Bash Manual

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